

PES University — Department of CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Software Engineering Laboratory Submission

Problem Statement #64

Disaster Relief Supply & Volunteer Coordinator

Category: Sustainability & Green Tech

An emergency disaster management platform mapping critical shelter supply needs (water, blankets, medical kits), dispatching volunteer skill sets, and tracking relief vehicle deliveries.

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1. Requirements Table

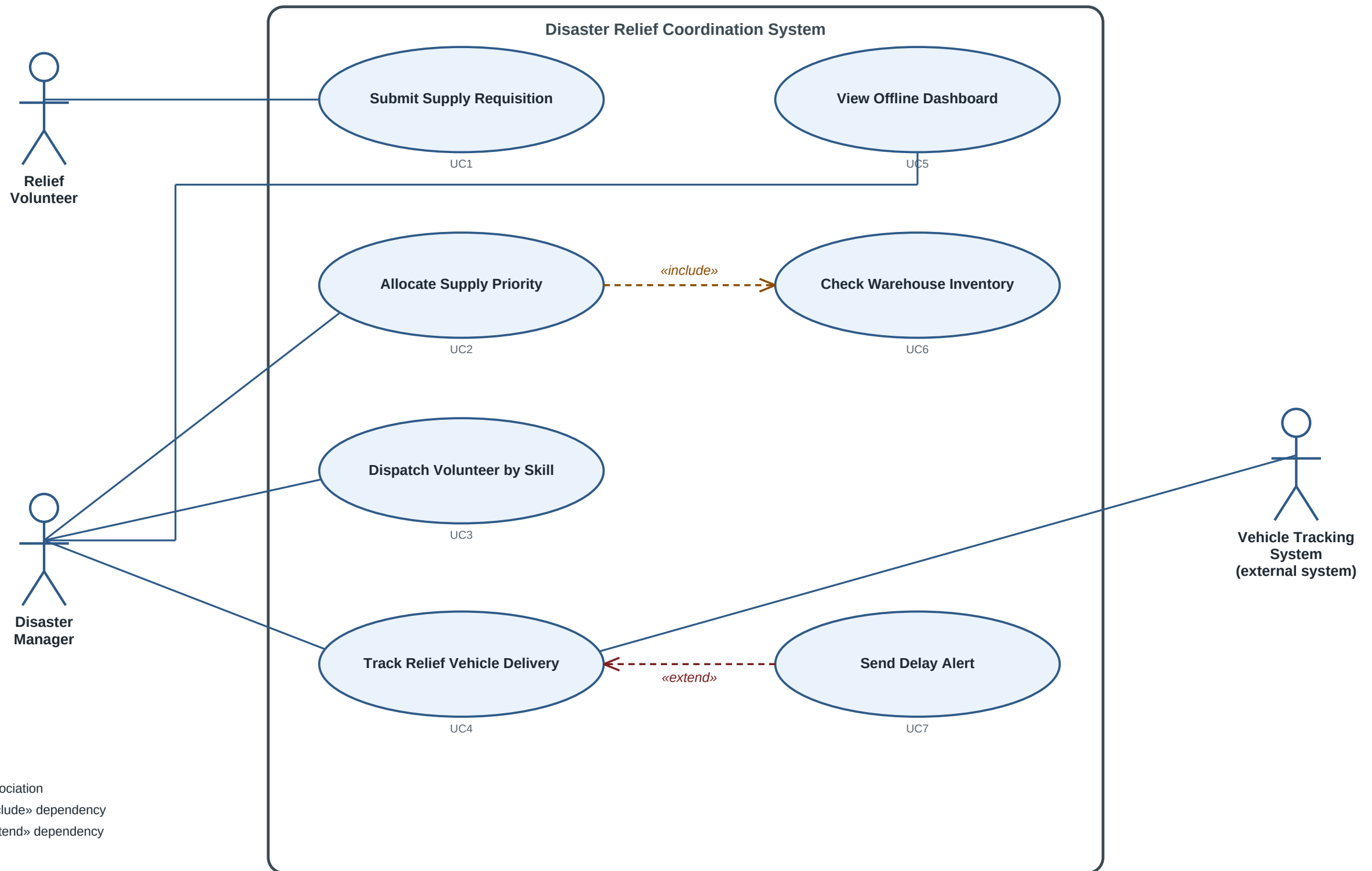
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	High	Pass: Critical need is given the highest dispatch priority and routed to the logistics manifest. Fail: A low-priority item is dispatched ahead of an outstanding water/medical requisition.	Core allocation functionality; ensures scarce relief supplies reach the most critical shelters first. (Provided by scenario.)
FR-002	Functional	The system shall allow a Relief Volunteer to submit a shelter supply requisition specifying item type, quantity, and urgency level from a mobile device.	High	Pass: A submitted requisition appears in the Disaster Manager's queue within 30 seconds and contains item type, quantity, and urgency. Fail: The requisition is missing a required field or does not appear within 30 seconds.	Accurate, timely field reporting is the entry point for all downstream supply-allocation decisions.
FR-003	Functional	The system shall allow the Disaster Manager to assign an available Relief Volunteer to a dispatch task based on the volunteer's registered skill set and current availability status.	High	Pass: Only volunteers whose skill tags match the task requirement and whose status is "Available" appear in the assignment list. Fail: A volunteer marked "Unavailable" or lacking the required skill is assigned.	Matching skills (medical, driving, heavy-lifting) to tasks prevents mis-deployment of scarce volunteer labour during an emergency.
FR-004	Functional	The system shall track the real-time GPS location of each relief vehicle and display an estimated delivery time on the Disaster Manager's dashboard.	Medium	Pass: Vehicle location updates on the map at least once every 60 seconds and the ETA is recalculated after each update. Fail: Vehicle position is stale for more than 60 seconds or no ETA is shown.	Visibility into vehicle location lets managers re-route deliveries around blocked roads and manage shelter expectations.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall automatically notify the Disaster Manager when a relief vehicle delivery is marked "Delivered" or "Failed."	Medium	Pass: A notification is received by the Disaster Manager within 15 seconds of the status change being recorded. Fail: No notification is sent, or it arrives more than 15 seconds after the status change.	Immediate delivery-outcome awareness allows managers to re-dispatch failed deliveries without waiting for a manual status check.
NFR-001	Nonfunctional (Performance & Security)	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load; the dashboard remains usable on a 2G-equivalent connection and cached map tiles load with no active connection. Fail: The dashboard fails to load or requires an active connection to view cached areas.	Disaster zones frequently have degraded or absent network infrastructure, so the system must remain operable offline. (Provided by scenario.)
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt all volunteer personal data and shelter location data both at rest and in transit using industry-standard encryption (e.g., AES-256 at rest, TLS 1.2+ in transit).	High	Pass: A security scan confirms all stored personal/location data is encrypted at rest and all network traffic is served over TLS 1.2 or higher, with no plaintext transmission detected. Fail: Any personal or location data is found unencrypted at rest or transmitted in plaintext.	Volunteer and shelter location data is sensitive; a breach could expose vulnerable populations and personnel to physical risk.

Note: FR-001 and NFR-001 (shaded rows) are the scenario-provided baseline requirements; their intended meaning has been preserved. FR-002 through FR-005 and NFR-002 were elicited from the Disaster Relief Supply & Volunteer Coordinator scenario.

UML Use-Case Diagram

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Use-Case Flow Specification

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Use Case ID	UC-02
Use Case Name	Allocate Supply Priority
Primary Actor	Disaster Manager
Related Requirement	FR-001 (priority allocation); includes UC-06 “Check Warehouse Inventory”

Preconditions

- The Disaster Manager is authenticated and logged into the coordination system.
- At least one pending shelter supply requisition (submitted via UC-01) exists in the system queue.

Postconditions

- The requisition is assigned a delivery priority rank and added to the logistics dispatch manifest.
- Warehouse stock levels are updated to reflect the reserved allocation.

Main Success Scenario

1. The Disaster Manager opens the “Pending Requisitions” queue.
2. The system displays all pending shelter requisitions with item type, quantity, urgency, and requesting shelter.
3. The Disaster Manager selects a requisition to process.
4. The system executes the included use case «include» “Check Warehouse Inventory” (UC-06) to retrieve current stock levels for the requested item.
5. The system compares the requisitioned quantity and urgency against available warehouse stock.
6. The system calculates and assigns a delivery priority (Critical / High / Medium / Low), giving water and medical kits precedence over other items at equal urgency.
7. The system adds the requisition to the logistics dispatch manifest at the assigned priority position.
8. The system reserves the allocated stock quantity against the warehouse inventory.
9. The Disaster Manager confirms the allocation.
10. The use case ends successfully.

Alternate Flow — 5a: Supply Shortage (Insufficient Warehouse Stock)

- 5a1. At step 5, the system determines that available warehouse stock is less than the requisitioned quantity.
- 5a2. The system flags the requisition as “Partially Fulfillable” and calculates the maximum quantity that can be allocated from current stock.
- 5a3. The system displays a shortage alert to the Disaster Manager, showing the shortfall amount.
- 5a4. The Disaster Manager either (a) approves a partial allocation, resuming at step 6 with the reduced quantity, or (b) escalates the requisition to a regional supply hub for replenishment.
- 5a5. If escalated, the system logs the requisition status as “Pending External Resupply” and the use case ends without completing allocation.